

Project Design Phase

Problem – Solution Fit Template

Date	30-06-2026
Team ID	23MH1A4257
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Solution Fit:

The Solution Fit for a credit card approval system bridges the gap between manual underwriting bottlenecks and the need for efficient risk management. By replacing slow, error-prone manual verification with an automated machine learning model, banks can instantly analyze complex applicant data like credit history, income, and debt ratios. This provides data-driven insights that eliminate subjective decision-making, allowing institutions to accurately filter out high-risk profiles while immediately approving eligible applicants.

Strategically, this automated approach improves the customer onboarding experience by eliminating lengthy wait times and vague approval processes. For applicants who don't meet the primary threshold, the system can automatically trigger alternative paths, such as routing them to credit-builder options or targeted financial products. This turns a high-friction process into a transparent, low-friction touchpoint that builds customer trust and reduces drop-off rates while maintaining tight regulatory and risk compliance.

Define CS, fill into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? In this case our customers are: - Banks and Financial Institutions - Credit Card Issuance Teams - Risk Analysts and Underwriters - End Customers (Credit Card Applicants) Primary users are banks to automate and improve the credit card approval process.	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices? - High rate of manual verification and lengthy approval process - Inconsistent decision making across different agents - Risk of approving high-risk applicants (default) - Lack of data-driven insights for decision making - Limited IT budget and resources for advanced solutions - Data quality issues (missing/inaccurate values) - Regulatory compliance and data privacy requirements	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What are the pros & cons? - Manual Verification: Reliable but slow, costly, inconsistent - Rule-based Systems: Fast but rigid, cannot capture complex patterns - Traditional Scoring Models: Better than rules but need expert tuning - Machine Learning Models (Our Solution): High accuracy, adapts to patterns, scalable - External Credit Bureaus: Useful but may be expensive or limited in coverage	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides. - Quickly and accurately decide whether to approve or reject a credit card application - Identify high-risk applicants to reduce default rate - Automate the decision-making process - Improve customer experience with faster approvals - Optimize risk vs reward (approve good customers, reject risky ones) - Reduce operational cost and manual effort	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? - Manual process is subjective and varies across agents - Lack of pattern recognition in large customer data - Existing rule-based systems are too rigid - Incomplete information about applicant's creditworthiness - Business wants to grow card base but avoid defaulters - No real-time data-driven risk assessment mechanism	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace) - Collects applicant details through online/offline forms - Verifies documents and income manually - Checks credit score from bureaus - Applies internal rules or scoring - Discusses borderline cases with senior officers - Takes final decision: Approve / Reject / Refer - Monitors defaults and updates policies periodically	Focus on J&P, tap into BE, understand RC
Identify strong TR & EM	3. TRIGGERS TR What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. - Customer applies for a credit card - Marketing campaigns / offers - Customer referrals - Need for higher credit limit - Existing card upgrade requests	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. Credit Card Approval Prediction System - Machine Learning model predicts probability of default based on applicant data - Data preprocessing handles missing values, encodes categories, scales features - Model trained on historical application data - Output: Approve / Reject / Refer with Risk Score - Dashboard for decision makers with insights - Continuously improves as more data is fed	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 - Bank website / mobile app application form - Internet banking portal - Customer support chat / email 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. - Branch visits - Paper application forms - Call center interactions	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? i.e. lost, insecure > confident, in control - use it in your communication strategy & design. Before: - Anxious, uncertain about approval, frustrated with long wait After: - Happy, satisfied with quick approval, confident in bank's process			

